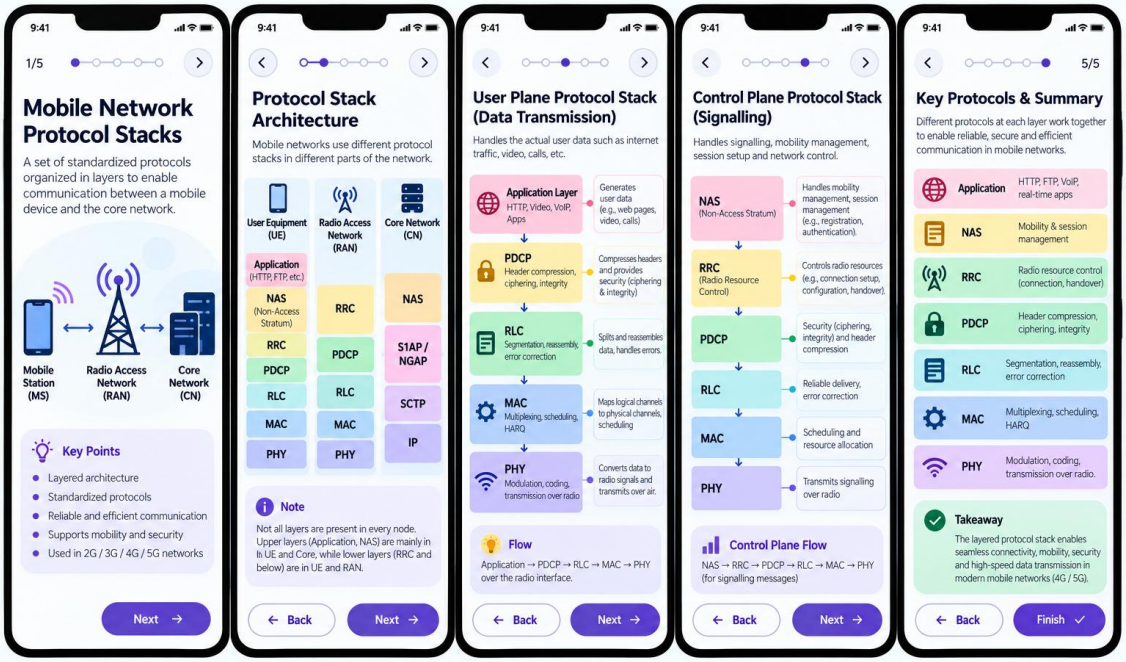

9:41
1/5
Mobile Network Protocol Stacks
A set of standardized protocols organized in layers to enable communication between a mobile device and the core network.
Mobile Station (MS)
Radio Access Network (RAN)
Core Network (CN)
Key Points
Layered architecture
Standardized protocols
Reliable and efficient communication
Supports mobility and security
Used in 2G / 3G / 4G / 5G networks
Next
9:41
Protocol Stack Architecture
Mobile networks use different protocol stacks in different parts of the network.
User Equipment (UE)
Application (HTTP, FTP, etc.)
NAS (Non-Access Stratum)
RRC
PDCP
RLC
MAC
PHY
Radio Access Network (RAN)
RRC
PDCP
RLC
MAC
PHY
Core Network (CN)
NAS
S1AP / NGAP
SCTP
IP
Note
Not all layers are present in every node. Upper layers (Application, NAS) are mainly in the UE and Core, while lower layers (RRC and below) are in UE and RAN.
Back
Next
9:41
User Plane Protocol Stack (Data Transmission)
Handles the actual user data such as internet traffic, video, calls, etc.
Application Layer
HTTP, Video, VoIP, Apps
Generates user data (e.g., web pages, video, calls)
PDCP
Header compression, ciphering, integrity
Compresses headers and provides security (ciphering & integrity)
RLC
Segmentation, reassembly, error correction
Splits and reassembles data, handles errors.
MAC
Multiplexing, scheduling, HARQ
Maps logical channels to physical channels, scheduling
PHY
Modulation, coding, transmission over radio
Converts data to radio signals and transmits over air.
Flow
Application → PDCP → RLC → MAC → PHY over the radio interface.
Back
Next
9:41
Control Plane Protocol Stack (Signalling)
Handles signalling, mobility management, session setup and network control.
NAS (Non-Access Stratum)
Handles mobility management, session management (e.g., registration, authentication).
RRC (Radio Resource Control)
Controls radio resources (e.g., connection setup, configuration, handover).
PDCP
Security (ciphering, integrity) and header compression
RLC
Reliable delivery, error correction
MAC
Scheduling and resource allocation
PHY
Transmits signalling over radio
Control Plane Flow
NAS → RRC → PDCP → RLC → MAC → PHY (for signalling messages)
Back
Next
9:41
5/5
Key Protocols & Summary
Different protocols at each layer work together to enable reliable, secure and efficient communication in mobile networks.
Application
HTTP, FTP, VoIP, real-time apps
NAS
Mobility & session management
RRC
Radio resource control (connection, handover)
PDCP
Header compression, ciphering, integrity
RLC
Segmentation, reassembly, error correction
MAC
Multiplexing, scheduling, HARQ
PHY
Modulation, coding, transmission over radio.
Takeaway
The layered protocol stack enables seamless connectivity, mobility, security and high-speed data transmission in modern mobile networks (4G / 5G).
Back
Finish